

Optimization of Neural Networks for Satellite Imagery-Based Disaster Risk Monitoring

SatRisk-Net *near real-time wildfire segmentation, on the edge.*

Supervisor: Asst. Prof. Dr. Selma Dilek

Emre Erdoğan · Emre Yeğin · Özgün Serergün Koca · Mustafa Ege · Kağan Canerik



DEMO VIDEO

01 Introduction · The Crisis

WHY THIS MATTERS



Wildfires can spread quickly and damage large areas, so affected regions must be identified as early as possible. Satellite imagery provides a powerful way to observe disasters from above, but large satellite images are difficult to process quickly with traditional cloud-based workflows. In this project, burned areas are segmented from satellite images using deep learning models. The trained models are then optimized for faster and lighter inference on edge devices such as NVIDIA Jetson Orin Nano.

Datasets · open, multi-region, human-verified

HLS Burn Scars

NASA · HLS · 6 bands · 512² tiles
Harmonized Landsat-Sentinel imagery with human-verified burn-scar masks. Our primary benchmark.

CEMS

Copernicus EMS · Sentinel-2
Real European emergency-mapping events. Tests cross-region generalization on operational data.

Burned Area

Public · Sentinel-2 · multispectral
Used for hyperparameter sensitivity analysis — small set, varied terrain, fast iteration.

3 × 3

Three segmentation networks — U-Net, DeepLabV3+, SegFormer — evaluated on **three independent wildfire datasets** (HLS Burn Scars, CEMS, Burned Area), then compressed to run live on an NVIDIA Jetson Orin Nano with TensorRT.

Why we moved inference to the edge

TRADITIONAL



EDGE AI · OURS



A traditional ground pipeline can take hours from satellite pass to actionable map. **Running models on edge devices near the data source** can reduce latency and bandwidth usage.

02 Model Training

THREE ARCHITECTURES · ONE PIPELINE

We benchmark **U-Net (ResNet-18)**, **DeepLabV3+ (ResNet-50)**, and **SegFormer-B0** under one shared pipeline so the architectures — not the plumbing — decide the winner. All three are adapted from RGB to **6-band HLS imagery** (B, G, R, NIR, SWIR-1, SWIR-2), trained with masked BCE / Dice / Focal losses, AdamW + AMP, and identical split protocols across every dataset.

STEP 1

Dataset

Sentinel-2 / HLS GeoTIFF · load · cloud-mask · locked test split

STEP 2

Preprocessing

Tile 512² · per-band 2~98% normalize · hflip + vflip + rot90 augment

STEP 3

Training

U-Net · DeepLabV3+ · SegFormer · AdamW + AMP · BCE + Dice masked · warmup + cosine · grad clip 1.0

STEP 4

Hyperparameter Search

LR · loss · grad-clip · augmentation · scheduler ablations

STEP 5

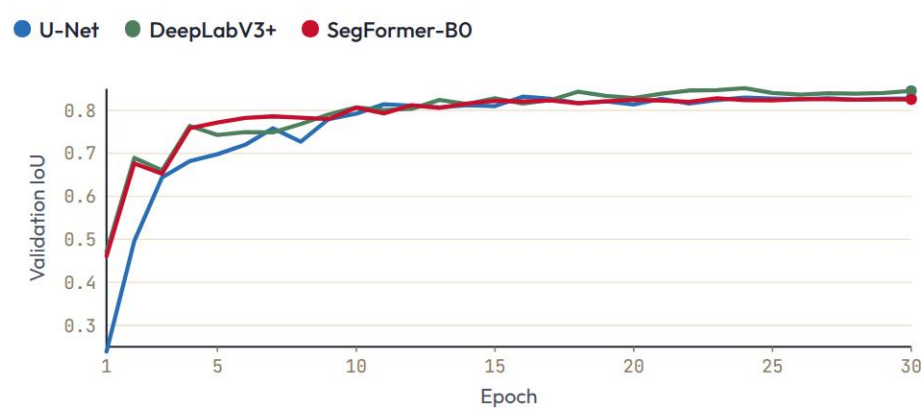
Validation & IoU Evaluation

SWA + TTA + Ensemble · val-tuned threshold · held-out test + TP/FP/FN overlays

Training strategy

- **One pipeline, fair comparison.** Same splits, same preprocessing, same metric — only the architecture changes.
- **Class imbalance is the hard part.** Disaster pixels are rare; we use Dice / Focal alongside BCE to keep the gradient honest.
- **Controlled ablations.** LR ∈ {5e-5, 1e-4, 2e-4}, gradient clipping, encoder freezing, AMP, scheduler choice — each varied independently.
- **Reproducibility first.** Locked seeds, locked HF test split, W&B-tracked runs, deterministic preprocessing.
- **30 epochs, best-val checkpoint.** Selected purely on validation IoU before touching the test set.

Validation IoU — HLS Burn Scars · 30 epochs



All three architectures converge into the 0.80 – 0.83 band on HLS. Transformer (SegFormer) leads on validation; CNNs catch up on test (see §04).

03 Results

THREE DATASETS · TEST SET IOU

Three segmentation models were evaluated on three wildfire datasets. Best architecture is dataset-dependent: **U-Net** leads on the smaller CEMS, **DeepLabV3+** on the larger HLS. SegFormer, which collapsed to 0.11 IoU under a naive from-scratch baseline on CEMS, was recovered to **≥ 0.80 across both datasets** through patch-embed surgery (3→12 ch) and transformer-appropriate fine-tuning.

BEST U-NET

0.8280

CEMS · best test IoU

HLS 0.8086

BEST DEEPLABV3+

0.8428

HLS · best test IoU

CEMS 0.7908

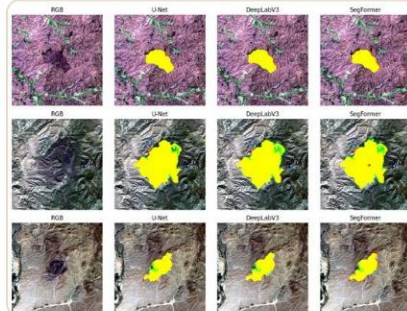
BEST SEGFORMER-B0

0.8146

HLS · best test IoU

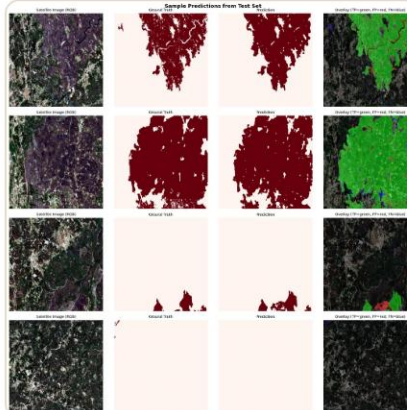
CEMS 0.7985

HLS Burn Scars · 3 architectures compared



Held-out scenes, side-by-side. All models recover the dominant burn scar; differences emerge at **edges and small detached patches**.

CEMS · U-Net test predictions



Sentinel-2 RGB · ground truth · U-Net prediction · overlay (TP green, FP red, FN blue). Reaches **0.8280 IoU** on the operational CEMS split.

Hyperparameter ablation · DeepLabV3+ on Burned Area (IoU, higher is better)

CHANGE APPLIED	IOU
★ Grad clip 1.0	0.662
Loss · Focal + Dice	0.634
Scheduler · ReduceOnPlateau	0.626
Freeze encoder (2 ep)	0.623
Baseline	0.620
LR = 5e-5	0.593

04 Edge Optimization

JETSON ORIN NANO · TENSORRT

Pruning was applied to make the models lighter and faster on Jetson.

Structured pruning removes channel-level parts of the network and gave the best accuracy–speed trade-off, especially for U-Net and DeepLabV3+.

How we measured both passes

- 1 **Host GPU.** Pruning + reduced precision compared against the unpruned PyTorch baseline.
- 2 **On-device.** Best variants exported via ONNX, compiled into a **TensorRT engine**, re-benchmarked on the Orin Nano at 7 W – MAXN.

KEY FINDING

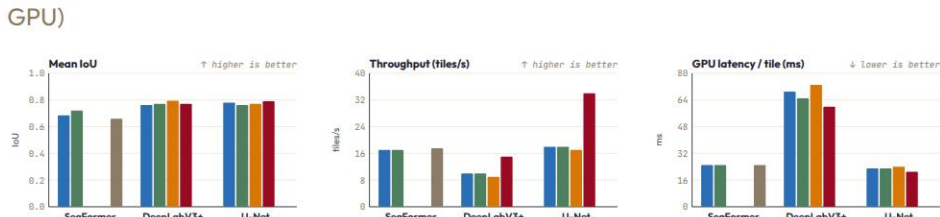
Structured pruning gave the best Jetson accuracy–speed trade-off.

NVIDIA Jetson Orin Nano

A compact **edge-AI device** for running neural networks locally. Reduces cloud dependency and supports faster inference near the data source.



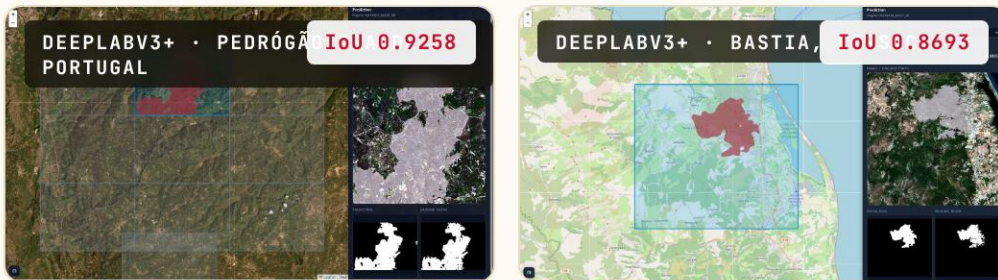
PyTorch · pruning & precision sweep (host GPU)



05 Interface & Future Vision

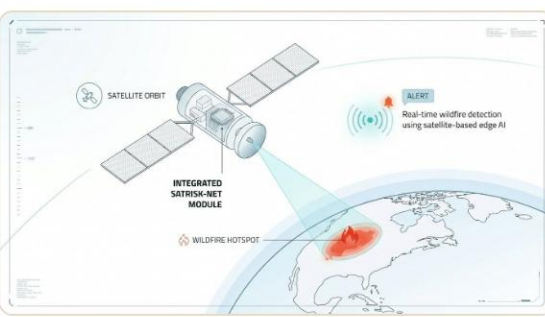
WEBSITE · WHAT'S NEXT

The trained models are wrapped in a **Leaflet + React web interface** so anyone can browse predictions on a world map, compare architectures side-by-side, toggle the ground truth overlay, and inspect the per-region IoU. The backend serves cached predictions for every Copernicus EMS event we evaluated.



What a user can do

- **Pick a region** from any indexed wildfire event.
- **Switch the model** — U-Net / DeepLabV3+ / SegFormer — and watch the prediction update.
- **Toggle ground-truth overlay** and read the per-region IoU live.
- **Compare segmentation outputs** across different architectures .



- A Sensor.** Sentinel-2 / Landsat captures multispectral imagery on every orbital pass.
- B Edge inference.** Running models on edge devices near the data source can reduce latency and bandwidth usage.
- C Near real-time alert.** Only compact georeferenced masks and alert payloads are transmitted, significantly reducing bandwidth compared to raw